

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400630
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Boesen; Peter Vincent

Selective audio isolation from body generated sound system and method

Abstract

A wireless earpiece includes a wireless earpiece housing, a processor disposed within the wireless earpiece housing, at least one microphone operatively connected to the processor, and at least one speaker operatively connected to the processor. The processor is configured to receive audio from the at least one microphone, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker. The processing of the audio involves identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds.

Inventors:	Boesen; Peter Vincent (Munich, DE)
Applicant:	BRAGI GmbH (Munich, DE)
Family ID:	1000008778576
Assignee:	BRAGI GmbH (Munich, DE)
Appl. No.:	18/436577
Filed:	February 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240185824 A1	Jun. 06, 2024

Related U.S. Application Data

continuation parent-doc US 17879319 20220802 US 11908442 child-doc US 18436577
continuation parent-doc US 17147329 20210112 US 11417307 20220816 child-doc US 17879319
continuation parent-doc US 16101894 20180813 US 10896665 20210119 child-doc US 17147329

Publication Classification

Int. Cl.: **G10K11/178** (20060101); **H04R1/10** (20060101)

U.S. Cl.:

CPC **G10K11/178** (20130101); **H04R1/1083** (20130101); G10K2210/1081 (20130101);
G10K2210/3028 (20130101); G10K2210/3045 (20130101); H04R1/1016 (20130101);
H04R1/1041 (20130101); H04R2201/107 (20130101)

Field of Classification Search

CPC: H04R (1/1041); H04R (1/1016); H04R (1/1083); G10K (11/178); G10K (2210/3045);
G10K (2210/3028); G10K (2210/1081)

USPC: 381/71.1; 381/71.6; 381/74; 381/94.1; 381/94.3

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2325590	12/1942	Carlisle et al.	N/A	N/A
2430229	12/1946	Kelsey	N/A	N/A
3047089	12/1961	Zwislocki	N/A	N/A
D208784	12/1966	Sanzone	N/A	N/A
3586794	12/1970	Michaelis	N/A	N/A
3696377	12/1971	Wall	N/A	N/A
3934100	12/1975	Harada	N/A	N/A
3983336	12/1975	Malek et al.	N/A	N/A
4069400	12/1977	Johanson et al.	N/A	N/A
4150262	12/1978	Ono	N/A	N/A
4334315	12/1981	Ono et al.	N/A	N/A
D266271	12/1981	Johanson et al.	N/A	N/A
4375016	12/1982	Harada	N/A	N/A
4588867	12/1985	Konomi	N/A	N/A
4617429	12/1985	Bellafiore	N/A	N/A
4654883	12/1986	Iwata	N/A	N/A
4682180	12/1986	Gans	N/A	N/A
4791673	12/1987	Schreiber	N/A	N/A
4852177	12/1988	Ambrose	N/A	N/A
4865044	12/1988	Wallace et al.	N/A	N/A
4984277	12/1990	Bisgaard et al.	N/A	N/A
5008943	12/1990	Arndt et al.	N/A	N/A
5185802	12/1992	Stanton	N/A	N/A
5191602	12/1992	Regen et al.	N/A	N/A
5201007	12/1992	Ward et al.	N/A	N/A

5201008	12/1992	Arndt et al.	N/A	N/A
D340286	12/1992	Seo	N/A	N/A
5280524	12/1993	Norris	N/A	N/A
5295193	12/1993	Ono	N/A	N/A
5298692	12/1993	Ikeda et al.	N/A	N/A
5343532	12/1993	Shugart	N/A	N/A
5347584	12/1993	Narisawa	N/A	N/A
5363444	12/1993	Norris	N/A	N/A
5444786	12/1994	Raviv	N/A	N/A
D367113	12/1995	Weeks	N/A	N/A
5497339	12/1995	Bernard	N/A	N/A
5606621	12/1996	Reiter et al.	N/A	N/A
5613222	12/1996	Guenther	N/A	N/A
5654530	12/1996	Sauer et al.	N/A	N/A
5689252	12/1996	Ayanoglu et al.	N/A	N/A
5692059	12/1996	Kruger	N/A	N/A
5721783	12/1997	Anderson	N/A	N/A
5748743	12/1997	Weeks	N/A	N/A
5749072	12/1997	Mazurkiewicz et al.	N/A	N/A
5771438	12/1997	Palermo et al.	N/A	N/A
D397796	12/1997	Yabe et al.	N/A	N/A
5802167	12/1997	Hong	N/A	N/A
5844996	12/1997	Enzmann et al.	N/A	N/A
D410008	12/1998	Almqvist	N/A	N/A
5929774	12/1998	Charlton	N/A	N/A
5933506	12/1998	Aoki et al.	N/A	N/A
5949896	12/1998	Nageno et al.	N/A	N/A
5987146	12/1998	Pluvinage et al.	N/A	N/A
6021207	12/1999	Puthuff et al.	N/A	N/A
6054989	12/1999	Robertson et al.	N/A	N/A
6081724	12/1999	Wilson	N/A	N/A
6084526	12/1999	Blotky et al.	N/A	N/A
6094492	12/1999	Boesen	N/A	N/A
6111569	12/1999	Brusky et al.	N/A	N/A
6112103	12/1999	Puthuff	N/A	N/A
6157727	12/1999	Rueda	N/A	N/A
6167039	12/1999	Karlsson et al.	N/A	N/A
6181801	12/2000	Puthuff et al.	N/A	N/A
6185152	12/2000	Shen	N/A	N/A
6208372	12/2000	Barraclough	N/A	N/A
6230029	12/2000	Yegiazaryan et al.	N/A	N/A
6275789	12/2000	Moser et al.	N/A	N/A
6339754	12/2001	Flanagan et al.	N/A	N/A
D455835	12/2001	Anderson et al.	N/A	N/A
6408081	12/2001	Boesen	N/A	N/A
6424820	12/2001	Burdick et al.	N/A	N/A
D464039	12/2001	Boesen	N/A	N/A
6470893	12/2001	Boesen	N/A	N/A
D468299	12/2002	Boesen	N/A	N/A
D468300	12/2002	Boesen	N/A	N/A

6522266	12/2002	Soehren et al.	N/A	N/A
6542721	12/2002	Boesen	N/A	N/A
6560468	12/2002	Boesen	N/A	N/A
6563301	12/2002	Gventer	N/A	N/A
6654721	12/2002	Handelman	N/A	N/A
6664713	12/2002	Boesen	N/A	N/A
6681176	12/2003	Funk et al.	N/A	N/A
6690807	12/2003	Meyer	N/A	N/A
6694180	12/2003	Boesen	N/A	N/A
6718043	12/2003	Boesen	N/A	N/A
6721657	12/2003	Ford et al.	N/A	N/A
6738485	12/2003	Boesen	N/A	N/A
6748095	12/2003	Goss	N/A	N/A
6754358	12/2003	Boesen et al.	N/A	N/A
6784873	12/2003	Boesen et al.	N/A	N/A
6807445	12/2003	Baumann	N/A	N/A
6823195	12/2003	Boesen	N/A	N/A
6852084	12/2004	Boesen	N/A	N/A
6879698	12/2004	Boesen	N/A	N/A
6892082	12/2004	Boesen	N/A	N/A
6920229	12/2004	Boesen	N/A	N/A
6952483	12/2004	Boesen et al.	N/A	N/A
6987986	12/2005	Boesen	N/A	N/A
7010137	12/2005	Leedom et al.	N/A	N/A
7113611	12/2005	Leedom et al.	N/A	N/A
D532520	12/2005	Kampmeier et al.	N/A	N/A
7136282	12/2005	Rebeske	N/A	N/A
7203331	12/2006	Boesen	N/A	N/A
7209569	12/2006	Boesen	N/A	N/A
7215790	12/2006	Boesen et al.	N/A	N/A
D549222	12/2006	Huang	N/A	N/A
D554756	12/2006	Sjursen et al.	N/A	N/A
7403629	12/2007	Aceti et al.	N/A	N/A
D579006	12/2007	Kim et al.	N/A	N/A
7447630	12/2007	Liu et al.	N/A	N/A
7463902	12/2007	Boesen	N/A	N/A
7508411	12/2008	Boesen	N/A	N/A
7532901	12/2008	LaFranchise et al.	N/A	N/A
7574008	12/2008	Zhang	N/A	N/A
D601134	12/2008	Elabidi et al.	N/A	N/A
7668652	12/2009	Spencer et al.	N/A	N/A
7825626	12/2009	Kozisek	N/A	N/A
7859469	12/2009	Rosener et al.	N/A	N/A
7965855	12/2010	Ham	N/A	N/A
7979035	12/2010	Griffin et al.	N/A	N/A
7983628	12/2010	Boesen	N/A	N/A
D647491	12/2010	Chen et al.	N/A	N/A
8095188	12/2011	Shi	N/A	N/A
8108143	12/2011	Tester	N/A	N/A
8140357	12/2011	Boesen	N/A	N/A

8238967	12/2011	Arnold et al.	N/A	N/A
8253589	12/2011	Grimm et al.	N/A	N/A
D666581	12/2011	Perez	N/A	N/A
8270637	12/2011	Abolfathi	N/A	N/A
8300864	12/2011	Müllenborn et al.	N/A	N/A
8383925	12/2012	Uehara	N/A	N/A
8406448	12/2012	Lin et al.	N/A	N/A
8430817	12/2012	Al-Ali et al.	N/A	N/A
8436780	12/2012	Schantz et al.	N/A	N/A
D687021	12/2012	Yuen	N/A	N/A
8483416	12/2012	Roeck	381/313	H04R 25/552
8675884	12/2013	Yehuday	N/A	N/A
8679012	12/2013	Kayyali	N/A	N/A
8719877	12/2013	VonDoenhoff et al.	N/A	N/A
8774434	12/2013	Zhao et al.	N/A	N/A
8831266	12/2013	Huang	N/A	N/A
8891800	12/2013	Shaffer	N/A	N/A
8994498	12/2014	Agrafioti et al.	N/A	N/A
D728107	12/2014	Martin et al.	N/A	N/A
9013145	12/2014	Castillo et al.	N/A	N/A
9037125	12/2014	Kadous	N/A	N/A
D733103	12/2014	Jeong et al.	N/A	N/A
9081944	12/2014	Camacho et al.	N/A	N/A
9229227	12/2015	Border et al.	N/A	N/A
9317241	12/2015	Tranchina	N/A	N/A
9414964	12/2015	Censo	N/A	H04R 1/1083
9461403	12/2015	Gao et al.	N/A	N/A
9510159	12/2015	Cuddihy et al.	N/A	N/A
D773439	12/2015	Walker	N/A	N/A
D775158	12/2015	Dong	N/A	N/A
9524631	12/2015	Agrawal et al.	N/A	N/A
D777710	12/2016	Palmborg	N/A	N/A
9544689	12/2016	Fisher et al.	N/A	N/A
D788079	12/2016	Son et al.	N/A	N/A
9684778	12/2016	Tharappel et al.	N/A	N/A
9711062	12/2016	Ellis et al.	N/A	N/A
9729957	12/2016	Kaller	N/A	H04R 3/02
9729979	12/2016	Özden	N/A	N/A
9767709	12/2016	Ellis	N/A	N/A
9786265	12/2016	Beckman	N/A	G10K 11/17857
9818005	12/2016	Yeager et al.	N/A	N/A
9821767	12/2016	Nixon	N/A	N/A
9848257	12/2016	Ambrose et al.	N/A	N/A
10056069	12/2017	Golani	N/A	A61F 5/56
10284968	12/2018	Van Dijk	N/A	H04R 25/305
2001/0005197	12/2000	Mishra et al.	N/A	N/A
2001/0027121	12/2000	Boesen	N/A	N/A
2001/0043707	12/2000	Leedom	N/A	N/A
2001/0056350	12/2000	Calderone et al.	N/A	N/A

2002/0002413	12/2001	Tokue	N/A	N/A
2002/0007510	12/2001	Mann	N/A	N/A
2002/0010590	12/2001	Lee	N/A	N/A
2002/0030637	12/2001	Mann	N/A	N/A
2002/0046035	12/2001	Kitahara et al.	N/A	N/A
2002/0057810	12/2001	Boesen	N/A	N/A
2002/0076073	12/2001	Taenzer et al.	N/A	N/A
2002/0118852	12/2001	Boesen	N/A	N/A
2003/0002705	12/2002	Boesen	N/A	N/A
2003/0065504	12/2002	Kraemer et al.	N/A	N/A
2003/0100331	12/2002	Dress et al.	N/A	N/A
2003/0104806	12/2002	Ruef et al.	N/A	N/A
2003/0115068	12/2002	Boesen	N/A	N/A
2003/0125096	12/2002	Boesen	N/A	N/A
2003/0218064	12/2002	Conner et al.	N/A	N/A
2004/0070564	12/2003	Dawson et al.	N/A	N/A
2004/0102931	12/2003	Ellis et al.	N/A	N/A
2004/0160511	12/2003	Boesen	N/A	N/A
2005/0017842	12/2004	Dematteo	N/A	N/A
2005/0043056	12/2004	Boesen	N/A	N/A
2005/0094839	12/2004	Gwee	N/A	N/A
2005/0125320	12/2004	Boesen	N/A	N/A
2005/0148883	12/2004	Boesen	N/A	N/A
2005/0165663	12/2004	Razumov	N/A	N/A
2005/0196009	12/2004	Boesen	N/A	N/A
2005/0197063	12/2004	White	N/A	N/A
2005/0212911	12/2004	Marvit et al.	N/A	N/A
2005/0251455	12/2004	Boesen	N/A	N/A
2005/0266876	12/2004	Boesen	N/A	N/A
2006/0029246	12/2005	Boesen	N/A	N/A
2006/0064037	12/2005	Shalon	N/A	N/A
2006/0073787	12/2005	Lair et al.	N/A	N/A
2006/0074671	12/2005	Farmaner et al.	N/A	N/A
2006/0074808	12/2005	Boesen	N/A	N/A
2006/0166715	12/2005	Engelen et al.	N/A	N/A
2006/0166716	12/2005	Seshadri et al.	N/A	N/A
2006/0220915	12/2005	Bauer	N/A	N/A
2006/0258412	12/2005	Liu	N/A	N/A
2007/0102009	12/2006	Wong et al.	N/A	N/A
2007/0230714	12/2006	Armstrong	N/A	N/A
2007/0239225	12/2006	Saringer	N/A	N/A
2007/0242834	12/2006	Coutinho et al.	N/A	N/A
2007/0247800	12/2006	Smith et al.	N/A	N/A
2007/0269785	12/2006	Yamanoi	N/A	N/A
2008/0013747	12/2007	Tran	N/A	N/A
2008/0076972	12/2007	Dorogusker et al.	N/A	N/A
2008/0090622	12/2007	Kim et al.	N/A	N/A
2008/0102424	12/2007	Holljes	N/A	N/A
2008/0146890	12/2007	LeBoeuf	N/A	N/A
2008/0187163	12/2007	Goldstein	N/A	N/A

2008/0215239	12/2007	Lee	N/A	N/A
2008/0253583	12/2007	Goldstein	N/A	N/A
2008/0254780	12/2007	Kuhl et al.	N/A	N/A
2008/0255430	12/2007	Alexandersson et al.	N/A	N/A
2008/0298606	12/2007	Johnson et al.	N/A	N/A
2008/0318518	12/2007	Coutinho et al.	N/A	N/A
2009/0003620	12/2008	McKillop et al.	N/A	N/A
2009/0008275	12/2008	Ferrari et al.	N/A	N/A
2009/0017881	12/2008	Madrigal	N/A	N/A
2009/0041313	12/2008	Brown	N/A	N/A
2009/0073070	12/2008	Rofougaran	N/A	N/A
2009/0097689	12/2008	Prest et al.	N/A	N/A
2009/0105548	12/2008	Bart	N/A	N/A
2009/0154739	12/2008	Zellner	N/A	N/A
2009/0182913	12/2008	Rosenblatt et al.	N/A	N/A
2009/0191920	12/2008	Regen et al.	N/A	N/A
2009/0226017	12/2008	Abolfathi et al.	N/A	N/A
2009/0240947	12/2008	Goyal et al.	N/A	N/A
2009/0245559	12/2008	Boltyenkov et al.	N/A	N/A
2009/0261114	12/2008	McGuire et al.	N/A	N/A
2009/0296965	12/2008	Kojima	N/A	N/A
2009/0296968	12/2008	Wu et al.	N/A	N/A
2009/0299215	12/2008	Zhang	N/A	N/A
2009/0303073	12/2008	Gilling et al.	N/A	N/A
2009/0304210	12/2008	Weisman	N/A	N/A
2010/0007805	12/2009	Vitito	N/A	N/A
2010/0033313	12/2009	Keady et al.	N/A	N/A
2010/0075631	12/2009	Black et al.	N/A	N/A
2010/0106356	12/2009	Trepagnier et al.	N/A	N/A
2010/0166206	12/2009	Macours	N/A	N/A
2010/0168075	12/2009	Dahlstrom et al.	N/A	N/A
2010/0203831	12/2009	Muth	N/A	N/A
2010/0210212	12/2009	Sato	N/A	N/A
2010/0285771	12/2009	Peabody	N/A	N/A
2010/0290636	12/2009	Mao et al.	N/A	N/A
2010/0320961	12/2009	Castillo et al.	N/A	N/A
2011/0018731	12/2010	Linsky et al.	N/A	N/A
2011/0029041	12/2010	Wiskerke	N/A	N/A
2011/0103609	12/2010	Pelland et al.	N/A	N/A
2011/0137141	12/2010	Razoumov	N/A	N/A
2011/0140844	12/2010	McGuire et al.	N/A	N/A
2011/0140956	12/2010	Henry et al.	N/A	N/A
2011/0239497	12/2010	McGuire et al.	N/A	N/A
2011/0286615	12/2010	Olodort et al.	N/A	N/A
2011/0293105	12/2010	Arie et al.	N/A	N/A
2012/0057740	12/2011	Rosal	N/A	N/A
2012/0155670	12/2011	Rutschman	N/A	N/A
2012/0159617	12/2011	Wu et al.	N/A	N/A
2012/0162891	12/2011	Tranchina et al.	N/A	N/A
2012/0163626	12/2011	Booij et al.	N/A	N/A

2012/0197737	12/2011	LeBoeuf et al.	N/A	N/A
2012/0235883	12/2011	Border et al.	N/A	N/A
2012/0309453	12/2011	Maguire	N/A	N/A
2013/0106454	12/2012	Liu et al.	N/A	N/A
2013/0154826	12/2012	Ratajczyk	N/A	N/A
2013/0178967	12/2012	Mentz	N/A	N/A
2013/0200999	12/2012	Spodak et al.	N/A	N/A
2013/0204617	12/2012	Kuo et al.	N/A	N/A
2013/0293494	12/2012	Reshef	N/A	N/A
2013/0316642	12/2012	Newham	N/A	N/A
2013/0343585	12/2012	Bennett et al.	N/A	N/A
2013/0346168	12/2012	Zhou et al.	N/A	N/A
2014/0002357	12/2013	Pombo et al.	N/A	N/A
2014/0004912	12/2013	Rajakarunanayake	N/A	N/A
2014/0010391	12/2013	Ek et al.	N/A	N/A
2014/0014697	12/2013	Schmierer et al.	N/A	N/A
2014/0020089	12/2013	Perini, II	N/A	N/A
2014/0072136	12/2013	Tenenbaum et al.	N/A	N/A
2014/0072146	12/2013	Itkin et al.	N/A	N/A
2014/0073429	12/2013	Meneses et al.	N/A	N/A
2014/0079257	12/2013	Ruwe et al.	N/A	N/A
2014/0106677	12/2013	Altman	N/A	N/A
2014/0122116	12/2013	Smythe	N/A	N/A
2014/0146973	12/2013	Liu et al.	N/A	N/A
2014/0153768	12/2013	Hagen et al.	N/A	N/A
2014/0163771	12/2013	Demeniuk	N/A	N/A
2014/0185828	12/2013	Helbling	N/A	N/A
2014/0219467	12/2013	Kurtz	N/A	N/A
2014/0222462	12/2013	Shakil et al.	N/A	N/A
2014/0235169	12/2013	Parkinson et al.	N/A	N/A
2014/0237518	12/2013	Liu	N/A	N/A
2014/0270227	12/2013	Swanson	N/A	N/A
2014/0270271	12/2013	Dehe et al.	N/A	N/A
2014/0276227	12/2013	Pérez	N/A	N/A
2014/0279889	12/2013	Luna	N/A	N/A
2014/0310595	12/2013	Acharya et al.	N/A	N/A
2014/0321682	12/2013	Kofod-Hansen et al.	N/A	N/A
2014/0335908	12/2013	Krisch et al.	N/A	N/A
2014/0348367	12/2013	Vavrus et al.	N/A	N/A
2015/0028996	12/2014	Agrafioti et al.	N/A	N/A
2015/0035643	12/2014	Kursun	N/A	N/A
2015/0036835	12/2014	Chen	N/A	N/A
2015/0043765	12/2014	Merks	N/A	N/A
2015/0056584	12/2014	Boulware et al.	N/A	N/A
2015/0110587	12/2014	Hori	N/A	N/A
2015/0124058	12/2014	Okpeva et al.	N/A	N/A
2015/0148989	12/2014	Cooper et al.	N/A	N/A
2015/0181356	12/2014	Krystek et al.	N/A	N/A
2015/0230022	12/2014	Sakai et al.	N/A	N/A
2015/0245127	12/2014	Shaffer	N/A	N/A

2015/0256949	12/2014	Vanpoucke et al.	N/A	N/A
2015/0264472	12/2014	Aase	N/A	N/A
2015/0264501	12/2014	Hu et al.	N/A	N/A
2015/0310720	12/2014	Gettings et al.	N/A	N/A
2015/0317565	12/2014	Li et al.	N/A	N/A
2015/0358751	12/2014	Deng et al.	N/A	N/A
2015/0359436	12/2014	Shim et al.	N/A	N/A
2015/0364058	12/2014	Lagree	N/A	N/A
2015/0373467	12/2014	Gelter	N/A	N/A
2015/0373474	12/2014	Kraft et al.	N/A	N/A
2015/0379251	12/2014	Komaki	N/A	N/A
2016/0033280	12/2015	Moore	N/A	N/A
2016/0034249	12/2015	Lee et al.	N/A	N/A
2016/0071526	12/2015	Wingate et al.	N/A	N/A
2016/0072558	12/2015	Hirsch et al.	N/A	N/A
2016/0073189	12/2015	Lindén et al.	N/A	N/A
2016/0094550	12/2015	Bradley et al.	N/A	N/A
2016/0100262	12/2015	Inagaki	N/A	N/A
2016/0112811	12/2015	Jensen	N/A	N/A
2016/0119737	12/2015	Mehnert et al.	N/A	N/A
2016/0124707	12/2015	Ermilov	N/A	N/A
2016/0125892	12/2015	Bowen et al.	N/A	N/A
2016/0140870	12/2015	Connor	N/A	N/A
2016/0142818	12/2015	Park	N/A	N/A
2016/0162259	12/2015	Zhao et al.	N/A	N/A
2016/0209691	12/2015	Yang et al.	N/A	N/A
2016/0226713	12/2015	Dellinger et al.	N/A	N/A
2016/0253994	12/2015	Panchapagesan et al.	N/A	N/A
2016/0324478	12/2015	Goldstein	N/A	N/A
2016/0345104	12/2015	Margalit	N/A	N/A
2016/0345107	12/2015	Van Dijk et al.	N/A	N/A
2016/0352818	12/2015	Han et al.	N/A	N/A
2016/0353196	12/2015	Baker et al.	N/A	N/A
2016/0360350	12/2015	Watson et al.	N/A	N/A
2017/0021257	12/2016	Gilbert	N/A	N/A
2017/0046503	12/2016	Cho	N/A	N/A
2017/0059152	12/2016	Hirsch et al.	N/A	N/A
2017/0060262	12/2016	Hviid et al.	N/A	N/A
2017/0060269	12/2016	Förstner et al.	N/A	N/A
2017/0061751	12/2016	Loermann et al.	N/A	N/A
2017/0061817	12/2016	May	N/A	N/A
2017/0062913	12/2016	Hirsch et al.	N/A	N/A
2017/0064426	12/2016	Hviid	N/A	N/A
2017/0064428	12/2016	Hirsch	N/A	N/A
2017/0064432	12/2016	Hviid et al.	N/A	N/A
2017/0064437	12/2016	Hviid et al.	N/A	N/A
2017/0065228	12/2016	Hirano	N/A	N/A
2017/0078780	12/2016	Qian et al.	N/A	N/A
2017/0078785	12/2016	Qian et al.	N/A	N/A
2017/0096065	12/2016	Katsuno et al.	N/A	N/A

2017/0100277	12/2016	Ke	N/A	N/A
2017/0108918	12/2016	Boesen	N/A	N/A
2017/0109131	12/2016	Boesen	N/A	N/A
2017/0110124	12/2016	Boesen et al.	N/A	N/A
2017/0110899	12/2016	Boesen	N/A	N/A
2017/0111723	12/2016	Boesen	N/A	N/A
2017/0111725	12/2016	Boesen et al.	N/A	N/A
2017/0111726	12/2016	Martin et al.	N/A	N/A
2017/0111740	12/2016	Hviid et al.	N/A	N/A
2017/0112671	12/2016	Goldstein	N/A	N/A
2017/0119318	12/2016	Shay et al.	N/A	N/A
2017/0127168	12/2016	Briggs et al.	N/A	N/A
2017/0131094	12/2016	Kulik	N/A	N/A
2017/0142511	12/2016	Dennis	N/A	N/A
2017/0146801	12/2016	Stempora	N/A	N/A
2017/0150920	12/2016	Chang et al.	N/A	N/A
2017/0151085	12/2016	Chang et al.	N/A	N/A
2017/0151447	12/2016	Boesen	N/A	N/A
2017/0151668	12/2016	Boesen	N/A	N/A
2017/0151918	12/2016	Boesen	N/A	N/A
2017/0151930	12/2016	Boesen	N/A	N/A
2017/0151957	12/2016	Boesen	N/A	N/A
2017/0151959	12/2016	Boesen	N/A	N/A
2017/0153114	12/2016	Boesen	N/A	N/A
2017/0153636	12/2016	Boesen	N/A	N/A
2017/0154532	12/2016	Boesen	N/A	N/A
2017/0155985	12/2016	Boesen	N/A	N/A
2017/0155992	12/2016	Perianu et al.	N/A	N/A
2017/0155993	12/2016	Boesen	N/A	N/A
2017/0155997	12/2016	Boesen	N/A	N/A
2017/0155998	12/2016	Boesen	N/A	N/A
2017/0156000	12/2016	Boesen	N/A	N/A
2017/0161017	12/2016	Chereau et al.	N/A	N/A
2017/0164890	12/2016	Leip et al.	N/A	N/A
2017/0178631	12/2016	Boesen	N/A	N/A
2017/0180842	12/2016	Boesen	N/A	N/A
2017/0180843	12/2016	Perianu et al.	N/A	N/A
2017/0180897	12/2016	Perianu	N/A	N/A
2017/0188127	12/2016	Perianu et al.	N/A	N/A
2017/0188132	12/2016	Hirsch et al.	N/A	N/A
2017/0193978	12/2016	Goldman	N/A	N/A
2017/0195829	12/2016	Belverato et al.	N/A	N/A
2017/0208393	12/2016	Boesen	N/A	N/A
2017/0214987	12/2016	Boesen	N/A	N/A
2017/0215016	12/2016	Dohmen et al.	N/A	N/A
2017/0230752	12/2016	Dohmen et al.	N/A	N/A
2017/0251295	12/2016	Pergament et al.	N/A	N/A
2017/0251933	12/2016	Braun et al.	N/A	N/A
2017/0257698	12/2016	Boesen et al.	N/A	N/A
2017/0258329	12/2016	Marsh	N/A	N/A

2017/0263236	12/2016	Boesen et al.	N/A	N/A
2017/0263376	12/2016	Verschueren et al.	N/A	N/A
2017/0266494	12/2016	Crankson et al.	N/A	N/A
2017/0273622	12/2016	Boesen	N/A	N/A
2017/0280257	12/2016	Gordon et al.	N/A	N/A
2017/0297430	12/2016	Hori et al.	N/A	N/A
2017/0301337	12/2016	Golani et al.	N/A	N/A
2017/0361213	12/2016	Goslin et al.	N/A	N/A
2017/0366233	12/2016	Hviid et al.	N/A	N/A
2018/0007994	12/2017	Boesen et al.	N/A	N/A
2018/0008194	12/2017	Boesen	N/A	N/A
2018/0008198	12/2017	Kingscott	N/A	N/A
2018/0009447	12/2017	Boesen et al.	N/A	N/A
2018/0011006	12/2017	Kingscott	N/A	N/A
2018/0011682	12/2017	Milevski et al.	N/A	N/A
2018/0011994	12/2017	Boesen	N/A	N/A
2018/0012228	12/2017	Milevski et al.	N/A	N/A
2018/0013195	12/2017	Hviid et al.	N/A	N/A
2018/0014102	12/2017	Hirsch et al.	N/A	N/A
2018/0014103	12/2017	Martin	N/A	N/A
2018/0014104	12/2017	Boesen et al.	N/A	N/A
2018/0014107	12/2017	Razouane et al.	N/A	N/A
2018/0014108	12/2017	Dragicevic et al.	N/A	N/A
2018/0014109	12/2017	Boesen	N/A	N/A
2018/0014113	12/2017	Boesen	N/A	N/A
2018/0014140	12/2017	Milevski et al.	N/A	N/A
2018/0014436	12/2017	Milevski	N/A	N/A
2018/0027342	12/2017	Meier	N/A	N/A
2018/0034951	12/2017	Boesen	N/A	N/A
2018/0040093	12/2017	Boesen	N/A	N/A
2018/0042501	12/2017	Adi et al.	N/A	N/A
2018/0056903	12/2017	Mullett	N/A	N/A
2018/0063626	12/2017	Pong et al.	N/A	N/A
2018/0125415	12/2017	Reed et al.	N/A	N/A
2019/0045288	12/2018	Walraevens et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
204244472	12/2014	CN	N/A
104683519	12/2014	CN	N/A
104837094	12/2014	CN	N/A
1469659	12/2003	EP	N/A
1017252	12/2005	EP	N/A
2903186	12/2014	EP	N/A
2074817	12/1980	GB	N/A
2508226	12/2013	GB	N/A
06292195	12/1997	JP	N/A
2008103925	12/2007	WO	N/A
2008113053	12/2007	WO	N/A

2007034371	12/2007	WO	N/A
2011001433	12/2010	WO	N/A
2012071127	12/2011	WO	N/A
2013134956	12/2012	WO	N/A
2014046602	12/2013	WO	N/A
2014043179	12/2013	WO	N/A
2015061633	12/2014	WO	N/A
2015110577	12/2014	WO	N/A
2015110587	12/2014	WO	N/A
2016032990	12/2015	WO	N/A
2016108230	12/2015	WO	N/A
2016187869	12/2015	WO	N/A

OTHER PUBLICATIONS

Akkermans, “Acoustic Ear Recognition for Person Identification”, Automatic Identification Advanced Technologies, 2005 pp. 219-223. cited by applicant

Alzahrani et al: “A Multi-Channel Opto-Electronic Sensor to Accurately Monitor Heart Rate against Motion Artefact during Exercise”, Sensors, vol. 15, No. 10, Oct. 12, 2015, pp. 25681-25702, XPO55334602, DOI: 10.3390/s151025681 the whole document. cited by applicant

Announcing the \$3,333,333 Stretch Goal (Feb. 24, 2014) pp. 1-14. cited by applicant

Ben Coxworth: “Graphene-based ink could enable low-cost, foldable electronics”, “Journal of Physical Chemistry Letters”, Northwestern University, (May 22, 2013), pp. 1-7. cited by applicant

Blain: “World's first graphene speaker already superior to Sennheiser MX400”, <http://www.gizmag.com/graphene-speaker-beats-sennheiser-mx400/31660>, (Apr. 15, 2014). cited by applicant

BMW, “BMW introduces BMW Connected—The personalized digital assistant”, “<http://bmwblog.com/2016/01/05/bmw-introduces-bmw-connected-the-personalized-digital-assistant>”, (Jan. 5, 2016). cited by applicant

Bragi is on Facebook (2014), pp. 1-51. cited by applicant

Bragi Update—Arrival of Prototype Chassis Parts—More People—Awesomeness (May 13, 2014), pp. 1-8. cited by applicant

Bragi Update—Chinese New Year, Design Verification, Charging Case, More People, Timeline(Mar. 6, 2015), pp. 1-18. cited by applicant

Bragi Update—First Sleeves From Prototype Tool—Software Development Kit (Jun. 5, 2014), pp. 1-8. cited by applicant

Bragi Update—Let's Get Ready to Rumble, A Lot to be Done Over Christmas (Dec. 22, 2014), pp. 1-18. cited by applicant

Bragi Update—Memories From April—Update on Progress (Sep. 16, 2014), pp. 1-15. cited by applicant

Bragi Update—Memories from May—Update on Progress—Sweet (Oct. 13, 2014), pp. 1-16. cited by applicant

Bragi Update—Memories From One Month Before Kickstarter—Update on Progress (Jul. 10, 2014), pp. 1-17. cited by applicant

Bragi Update—Memories From The First Month of Kickstarter—Update on Progress (Aug. 1, 2014), pp. 1-16. cited by applicant

Bragi Update—Memories From The Second Month of Kickstarter—Update on Progress (Aug. 22, 2014), pp. 1-15. cited by applicant

Bragi Update—New People @Bragi—Prototypes (Jun. 26, 2014), pp. 1-9. cited by applicant

Bragi Update—Office Tour, Tour to China, Tour to CES (Dec. 11, 2014), pp. 1-14. cited by applicant

Bragi Update—Status on Wireless, Bits and Pieces, Testing—Oh Yeah, Timeline(Apr. 24, 2015), pp. 1-18. cited by applicant

Bragi Update—The App Preview, The Charger, The SDK, Bragi Funding and Chinese New Year (Feb. 11, 2015), pp. 1-19. cited by applicant

Bragi Update—What We Did Over Christmas, Las Vegas & CES (Jan. 19, 2014), pp. 1-21. cited by applicant

Bragi Update—Years of Development, Moments of Utter Joy and Finishing What We Started(Jun. 5, 2015), pp. 1-21. cited by applicant

Bragi Update—Alpha 5 and Back to China, Backer Day, on Track(May 16, 2015), pp. 1-15. cited by applicant

Bragi Update—Beta2 Production and Factory Line(Aug. 20, 2015), pp. 1-16. cited by applicant

Bragi Update—Certifications, Production, Ramping Up (Nov. 13, 2015), pp. 1-15. cited by applicant

Bragi Update—Developer Units Shipping and Status(Oct. 5, 2015), pp. 1-20. cited by applicant

Bragi Update—Developer Units Started Shipping and Status (Oct. 19, 2015), pp. 1-20. cited by applicant

Bragi Update—Developer Units, Investment, Story and Status(Nov. 2, 2015), pp. 1-14. cited by applicant

Bragi Update—Getting Close(Aug. 6, 2015), pp. 1-20. cited by applicant

Bragi Update—On Track, Design Verification, How It Works and What's Next(Jul. 15, 2015), pp. 1-17. cited by applicant

Bragi Update—On Track, On Track and Gems Overview (Jun. 24, 15), pp. 1-19. cited by applicant

Bragi Update—Status on Wireless, Supply, Timeline and Open House@Bragi(Apr. 1, 2015), pp. 1-17. cited by applicant

Bragi Update—Unpacking Video, Reviews on Audio Perform and Boy are we Getting Close(Sep. 10, 2015), pp. 1-15. cited by applicant

Farr, Christina: “iPads In Every Hospital: Apple's Plan to Crack The \$3 Trillion Health Care Sector”, “<https://www.fastcompany.com/3069060/artists-and-scientists-are-teaming-with-businesses-and-non-profits-on-gender-concerns>” (Mar. 18, 2017). cited by applicant

Healthcare Risk Management Review, “Nuance updates computer-assisted physician documentation solution” (Oct. 20, 2016), pp. 1-2. cited by applicant

Hoffman, “How to Use Android Beam to Wirelessly Transfer Content Between Devices”, (Feb. 22, 2013). cited by applicant

Hoyt et al., “Lessons Learned from Implementation of Voice Recognition for Documentation in the Military Electronic Health Record System”, The American Health Information Management Association (2017), pp. 1-8. cited by applicant

Hyundai Motor America, “Hyundai Motor Company Introduces A Health + Mobility Concept for Wellness in Mobility”, Fountain Valley, California (2017), pp. 1-3. cited by applicant

International Search Report & Written Opinion, PCT/EP2016/070216 (Oct. 18, 2016) 13 pages. cited by applicant

International Search Report & Written Opinion, PCT/EP2016/070231 (Nov. 18, 2016) 12 pages. cited by applicant

International Search Report & Written Opinion, PCT/EP2016/070245 (Nov. 16, 2016) 10 pages. cited by applicant

International Search Report & Written Opinion, PCT/EP2016/070247 (Nov. 18, 2016) 13 pages. cited by applicant

International Search Report and Written Opinion, PCT/EP2016/070228 (Jan. 9, 2017) 13 pages. cited by applicant

Jain A et al: “Score normalization in multimodal biometric systems”, Pattern Recognition, Elsevier, GB, vol. 38, No. 12, Dec. 31, 2005, pp. 2270-2285, XPO27610849, ISSN: 0031-3203. cited by

applicant

Last Push Before The Kickstarter Campaign Ends on Monday 4pm CET (Mar. 28, 2014), pp. 1-7. cited by applicant

Lovejoy: “Touch ID built into iPhone display one step closer as third-party company announces new tech”, “<http://9to5mac.com/2015/07/21/virtualhomebutton/>” (Jul. 21, 2015). cited by applicant

Nemanja Paunovic et al, “A methodology for testing complex professional electronic systems”, Serbian Journal of Electrical Engineering, vol. 9, No. 1, Feb. 1, 2012, pp. 71-80, XPO55317584, Yu. cited by applicant

Nigel Whitfield: “Fake tape detectors, ‘from the stands’ footie and Ugh? Internet of Things in my set-top box”; http://www.theregister.co.uk/2014/09/24/ibc_round_up_object_audio_dlna_iot/ (Sep. 24, 2014). cited by applicant

Nuance, “ING Netherlands Launches Voice Biometrics Payment System in the Mobile Banking App Powered by Nuance”, “<https://www.nuance.com/about-us/newsroom/press-releases/ing-netherlands-launches-nuance-voice-biometrics.html>”, 4 pages (Jul. 28, 2015). cited by applicant

Staab, Wayne J., et al., “A One-Size Disposable Hearing Aid is Introduced”, The Hearing Journal 53(4):36-41) Apr. 2000. cited by applicant

Stretchgoal—It's Your Dash (Feb. 14, 2014), pp. 1-14. cited by applicant

Stretchgoal—The Carrying Case for The Dash (Feb. 12, 2014), pp. 1-9. cited by applicant

Stretchgoal—Windows Phone Support (Feb. 17, 2014), pp. 1-17. cited by applicant

The Dash + The Charging Case & The Bragi News (Feb. 21, 2014), pp. 1-12. cited by applicant

The Dash—A Word From Our Software, Mechanical and Acoustics Team + An Update (Mar. 11, 2014), pp. 1-7. cited by applicant

Update From Bragi—\$3,000,000—Yippee (Mar. 22, 2014), pp. 1-11. cited by applicant

Weisiger; “Conjugated Hyperbilirubinemia”, Jan. 5, 2016. cited by applicant

Wertzner et al., “Analysis of fundamental frequency, jitter, shimmer and vocal intensity in children with phonological disorders”, V. 71, n.5, 582-588, Sep./Oct. 2005; Brazilian Journal of Othrrhinolaryngology. cited by applicant

Wikipedia, “Gamebook”, <https://en.wikipedia.org/wiki/Gamebook>, Sep. 3, 2017, 5 pages. cited by applicant

Wikipedia, “Kinect”, “<https://en.wikipedia.org/wiki/Kinect>”, 18 pages, (Sep. 9, 2017). cited by applicant

Wikipedia, “Wii Balance Board”, “https://en.wikipedia.org/wiki/Wii_Balance_Board”, 3 pages, (Jul. 20, 2017). cited by applicant

Primary Examiner: Paul; Disler

Attorney, Agent or Firm: Goodhue, Coleman & Owens, P.C.

Background/Summary

PRIORITY STATEMENT (1) This application is a continuation of U.S. patent application Ser. No. 17/879,319 filed Aug. 2, 2022, which is a continuation of U.S. patent application Ser. No. 17/147,329, filed Jan. 12, 2021 now U.S. Pat. No. 11,417,307, which is a continuation of U.S. patent application Ser. No. 16/101,894, filed Aug. 13, 2018 now U.S. Pat. No. 10,896,665, which is a continuation of U.S. patent application Ser. No. 15/801,045 filed on Nov. 1, 2017 now U.S. Pat. No. 10,062,373, which claims priority to U.S. Provisional Patent Application 62/417,195, filed on Nov. 3, 2016, all of which are titled Selective Audio Isolation from Body Generated Sound System and Method, and all of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention relates to wearable devices. More particularly, but not exclusively, the present invention relates to earpieces.

BACKGROUND

(2) People generate a great deal of body noise through their everyday movements. While such sounds are generally not a problem, when a user wears an earpiece, the user tends to focus on such sounds to the detriment of their use and enjoyment of the earpiece. What is needed is a system and method to reduce or eliminate such body noise, so a user may enjoy the use and enjoyment of the earpiece without undue distraction.

SUMMARY

(3) Therefore, it is a primary object, feature, or advantage of the present invention to improve over the state of the art.

(4) It is a further object, feature, or advantage of the present invention to eliminate body noise through the use of an active noise cancellation system.

(5) It is a still further object, feature, or advantage of the present invention to eliminate body noise through the filtering of body noise acquired using one or more microphones connected to an earpiece.

(6) Another object, feature, or advantage is to eliminate outside noise in addition to body noise.

(7) In one implementation, a system includes an earpiece having an earpiece housing, at least one microphone mounted onto the earpiece housing, wherein at least one microphone is configured to receive at least one body generated sound, a processor disposed within the earpiece housing and operatively connected to each microphone, wherein the processor is configured to neutralize each body generated sound, and a speaker operatively connected to the processor, wherein the speaker is configured to transduce audio signals.

(8) One or more of the following features may be included. The earpiece housing may be composed of soundproof materials. The earpiece may comprise a set of earpieces. The set of earpieces may comprise a left earpiece and a right earpiece. Each earpiece may be configured to substantially encompass an external auditory canal of a user. Each earpiece may be further configured to substantially fit within a user's ear canal. One or more microphones may comprise an air microphone and a bone conduction microphone. One or more microphones may be further configured to filter one or more body generated sounds. The body generated sound filtering may be performed within a specific decibel range. The neutralization of each body generated sound may be performed by superimposing a cancellation signal to an audio signal communicated by the speaker, wherein the cancellation signal is configured so the superimposition of each cancellation signal with each body generated noise substantially nets to zero decibels. The neutralization of each body generated sound may be performed by filtering each body generated sound from audio signals using one or more microphones. Audio signals to the speaker may be suppressed or attenuated if the audio signal exceeds a specific decibel range.

(9) In another implementation, a method of filtering body generated sounds from an audio signal using an earpiece includes providing at least one audio signal, receiving at least one body generated sound through a microphone, separating each body generated sound from each audio signal using a processor, and communicating each audio signal using a speaker.

(10) One or more of the following features may be included. One or more earpieces may be configured to substantially encompass an external auditory canal of a user. One or more microphones may filter one or more audio signals. The separation of each body generated sound from each audio signal may be performed through the generation of a cancellation signal by the processor, wherein the cancellation signal is configured as to substantially net to zero decibels when superimposed onto a body generated sound. The separation of each body generated sound from each audio signal may be performed through the filtering of each body generation sound.

(11) According to another aspect, a wireless earpiece includes a wireless earpiece housing, a processor disposed within the wireless earpiece housing, at least one microphone operatively connected to the processor, and at least one speaker operatively connected to the processor. The processor is configured to receive audio from the at least one microphone, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker. The processing of the audio may include identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds. The at least one microphone may include both an air microphone and a bone microphone. The identifying the body generated sounds may be performed by comparing a first audio signal from the air microphone with a second audio signal from the bone microphone. The removing the body generated sounds may be performed by superimposing a cancellation signal to the audio. The removing the body generated sounds may be performed by filtering the body generated sounds from the audio. The body generated sounds may include at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

(12) According to another aspect, a method of removing body generated sounds by an earpiece may include providing the earpiece. The earpiece may include a wireless earpiece housing, a processor disposed within the wireless earpiece housing, at least one microphone operatively connected to the processor and at least one speaker operatively connected to the processor, wherein the processor is configured to receive audio from the at least one microphone, perform processing on the audio to provide processed audio, and output the processed audio to the at least one speaker. The method may include receiving audio from the at least one microphone at the processor, performing processing on the audio to provide processed audio, wherein the processing of the audio comprises identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds, and outputting the processed audio to the at least one speaker of the wireless earpiece. The at least one microphone may include an air microphone and a bone microphone. The step of identifying the body generated sounds may be performed by comparing a first audio signal from the air microphone with a second audio signal from the bone microphone. The step of removing the body generated sounds may be performed using active noise cancellation. The step of removing the body generated sounds may be performed by filtering.

(13) According to another aspect, a set of wireless earpieces includes a first wireless earpiece having a first wireless earpiece housing, a processor disposed within the first wireless earpiece housing, at least one first wireless earpiece microphone operatively connected to the first wireless earpiece processor, a first wireless earpiece transceiver operatively connected to the first wireless earpiece processor, and at least one first wireless earpiece speaker operatively connected to the first wireless earpiece processor. The set of wireless earpieces may further include a second wireless earpiece comprising a second wireless earpiece housing different from the first wireless earpiece housing, at least one second wireless earpiece microphone, at least one second wireless earpiece speaker, and at least one second wireless earpiece transceiver. The transceiver of the second wireless earpiece may communicate with the transceiver of the first wireless earpiece. The first wireless earpiece processor may be configured to receive audio from the at least one first wireless earpiece microphone and from the at least one second wireless earpiece microphone, perform processing of the audio to provide processed audio, and output the processed audio to the at least one first wireless earpiece speaker. The processing of the audio may involve identifying body generated sounds generated by a body of a user of the first wireless earpiece and the second wireless earpiece and removing the body generated sounds. The at least one first wireless earpiece microphone may include a first wireless earpiece air microphone and a first wireless earpiece bone microphone. The at least one second wireless earpiece microphone may include a second wireless earpiece air microphone and a second wireless earpiece bone microphone. The step of identifying the body generated sounds may be performed by comparing audio signals from the first wireless earpiece air microphone with the first wireless earpiece bone microphone and from the second

wireless earpiece air microphone with the second wireless earpiece bone microphone. The step of removing the body generated sounds may be performed by superimposing a cancellation signal to the audio. The step of removing the body generated sounds may be performed by filtering the body generated sounds from the audio. The body generated sounds may include at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

(14) One or more of these and/or other objects, features, or advantages of the present invention will become apparent from the specification and following claims. No single embodiment need provide each and every object, feature, or advantage. Different embodiments may have different objects, features, or advantages. Therefore, the present invention is not to be limited to or by an object, feature, or advantage stated herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram regarding a first embodiment of the system.
- (2) FIG. 2 illustrates a left earpiece and a right earpiece.
- (3) FIG. 3 illustrates a side view of the right earpiece and its relationship with a user's ear.
- (4) FIG. 4 is another block diagram regarding a second embodiment of the system.
- (5) FIG. 5 includes a flowchart of one implementation of the method of filtering body generated sounds from an audio signal using an earpiece.
- (6) FIG. 6 includes a flowchart of a second implementation of the method of filtering body generated sounds from an audio signal using an earpiece.

DETAILED DESCRIPTION

(7) FIG. 1 shows a block diagram of one embodiment of the system **10** having an earpiece **12** with an earpiece housing **14**, one or more microphones **16** associated with the earpiece housing **14**, a processor **18** disposed within the earpiece housing **14** and operatively connected to each microphone **16**, and a speaker **20** disposed within the earpiece housing **14** and operatively connected to the processor **18**. The earpiece housing **14** may be composed of any material resistant to shear and strain and unlikely to cause skin allergies or rashes. In addition, the earpiece housing **14** may be composed of soundproof materials and may also be configured as to be waterproof. The earpiece housing **14** may also substantially encompass the external auditory canal of a user in order to substantially reduce or eliminate external sounds. One or more microphones **16** associated with the external housing **14** may be configured to receive external sounds in addition to body generated sounds.

(8) One or more microphones **16** may be located on or within any part of the earpiece housing **14** conducive to receiving a body generated sound and may also include air and bone conduction microphones configured to receive sounds. In one mode of operation, the wireless earpiece may prevent environmental sounds from being heard by a user. In another mode of operation, the wireless earpiece **12** may provide an audio transparency function where sounds detected from the environment with one or more microphones **16** may be reproduced at one or more speakers **20** of the wireless earpiece **12**.

(9) Various methods of identifying body generated sounds are contemplated. In one embodiment audio signals from one or more bone conduction microphones are compared with audio signals from one or more air conduction microphones to identify body generated sounds. In another embodiment, processing of audio signals may be used to determine a source of the sound, for example when two microphones of a wireless earpiece detect the same sound at different instances in time, a determination may be made if the sound propagated from the environment to the microphones or from the body to the microphones. In addition, body generated sounds may be determined by comparison of audio to known body generated sounds or comparison of properties

of audio signals to properties of known body generated sounds. Such comparisons may be facilitated through the use of training sets of body generated sounds or asking a user to generate different body generated sounds. Any number of other methods may be used to identify as sound as a body generated sound. The processor **18** is disposed within the earpiece housing **14** and operatively connected to each microphone **16** and is configured to neutralize or otherwise attenuate or cancel any body generated sound signal it receives. The processor **18** may perform this function through active noise cancellation, where a cancellation signal is superimposed onto each body generated sound through destructive interference so as to substantially reduce the effect of the body generated sound, or may simply filter out a body generated sound using an algorithm stored within the processor **18** configured to identify which sounds may be body generated sounds identified by the algorithm from reaching the processor **18**. These processes may be performed continuously or discretely, and the processor **18** may determine more than one cancellation signal if necessary depending on the body generated sounds received by the microphones **16**. The processor **18** may also be configured to produce the audio signals communicated to a user's tympanic membrane, filter out external sounds which are not body generated sounds, and/or regulate other functions of the earpiece **12**. The speaker **20** configured to transduce audio signals substantially free of body generated sounds to a user's tympanic membrane.

(10) FIG. 2 illustrates a system **10** which includes a left earpiece **13A** and a right earpiece **13B**. The left earpiece **13A** has a left earpiece housing **14A**. The right earpiece **13B** has a right earpiece housing **14B**, the left earpiece housing **14A** different from the right earpiece housing **14**. The left earpiece **13A** and the right earpiece **13B** may be configured to fit around a user's ear canal so as to minimize the amount of external sound capable of reaching the ear canal as well as configured to fit within the ear canal so as to minimize the distance between the speakers and a user's tympanic membranes and maintain good fit. The earpiece housings **14A** and **14B** may be composed of soundproof or shear resistant materials and may also be configured to be waterproof. A microphone **16A** is shown on the left earpiece **13A** and a microphone **16B** is shown on the right earpiece **13B**. The microphones **16A** and **16B** may be located anywhere on the earpieces **13A** and **13B** respectively and may be configured to receive ambient environmental sounds or a user's instructions to control one or more earpiece functions in addition to non-vocal body generated sounds. The earpieces **13A** and **13B** may be configured to communicate audio signals **40A** and **40B** respectively to a user's tympanic membranes.

(11) FIG. 3 illustrates a side view of a right earpiece **13B** and its relationship to a user's ear. The right earpiece **13B** may be configured to both minimize the amount of external sound reaching the user's ear canal **42** and to facilitate the transmission of the audio signal **40B** from the speaker **20** to a user's tympanic membrane **44**. The right earpiece **13B** may be configured to be of any size necessary to fit within the user's ear canal **42** and the distance between the speaker **20** and the user's tympanic membrane **44** may be any distance sufficient to facilitate transmission of the audio signal **40B** to the user's tympanic membrane **44**. A microphone **16** is shown on the outside of the right earpiece **13B**. The microphone **16** may be configured to, in addition to receiving body generated noise, receive sounds from a user or the outside environment which may be used to reconfigure, change, or otherwise modify one or more functions of the earpiece **13B**.

(12) FIG. 4 is a block diagram of another embodiment of the system which includes an earpiece **12** with an earpiece housing **14**, a plurality of sensors which may include an air microphone **16** and a bone microphone **34**. Additional microphones may be present. A speaker **20** is operatively connected to the processor **18**. Other examples of sensors **22** which may be present may include one or more inertial sensors **36**. A radio transceiver **26** is operatively connected to the one or more processors **18**. Another transceiver **24** may be present which may provide for communicating between left and right earpieces. The transceiver **24** may be a near field magnetic induction (NFMI) transceiver. One or more LEDs may also be operatively connected to the one or more processors **18** to provide for visual feedback of operations of the device.

(13) FIG. 5 illustrates one implementation of the method of removing body generated sounds. In step **102**, an audio signal is provided such as may be detected using one or more microphones of an earpiece. In step **104**, at least one body generated sound is identifying within the one or more audio signals. The body generated sounds may originate from anywhere on a user's body and need not be physiologic in nature. For example, a body generated sound may be due to a necklace moving around the user's neck, a sole installed within a shoe the user is wearing, baggy pants the user may be wearing, or otherwise be body generated. In step **106**, a processor separates any audio signals from body generated sounds. The process by which a processor may perform this step include through a noise cancellation signal or through filtering or active filtering using one or more algorithms or otherwise. In step **108**, the modified audio signal is transduced.

(14) FIG. 6 shows another example of a method of filtering body generated sounds from an audio signal using an earpiece. In step **202**, an earpiece is provided such as of one of the examples previously described. In step **204**, audio is received from one or more of the microphones. In step **206**, the audio is processed to remove at least one body generated sound. In step **212**, the resulting audio signal(s) are transduced at one or more speakers of the wireless earpiece.

(15) It should also be appreciated where there is a set of wireless earpieces such as shown in FIG. 2, audio from one of the wireless earpieces may communicated to the other earpiece and be processed by the other earpiece to remove body generated sounds. Having audio from both locations may provide additional information about the sound to assist in processing, for example, it may provide additional information regarding locating the sound.

(16) Thus, the system and method described herein allow for selectively monitoring for body generated sounds, e.g. the click or pop generated by movement of the user's temporomandibular joint. Such sounds may be detected by microphones incorporated into the earpiece device and then extracted from the audio output signal delivered through the speaker system of the device to the middle ear. Such a device allows for mitigating the effects of body induced sound in order to allow the user to concentrate more fully on the selected device output. Moreover, an earpiece as described herein allows for monitoring body created sound inputs delivered to the middle ear through the air or bone conduction systems present in the body, allows for creating effective audio isolation through the use of isolation measures which would include masking noise or signal extraction algorithms, and allows a user to more fully concentrate on the audio input delivered to the middle ear by reducing or subtracting the sound inputs from body generated noise inputs.

(17) Therefore, various methods, system, and apparatus have been shown and described. Although specific examples or embodiments are shown herein it is to be understood various elements or steps from different embodiments may be combined. It is to be further understood numerous options, variations, and alternatives are contemplated.

Claims

1. A wireless earpiece comprising: a wireless earpiece housing; a processor disposed within the wireless earpiece housing; a plurality of microphones operatively connected to the processor; at least one speaker operatively connected to the processor; wherein the processor is configured to receive audio from the plurality of microphones, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker to provide an audio transparency function; wherein the processing of the audio comprises identifying sounds to remove based on training data associated with sounds to remove and then remove the sounds to remove from the processed audio, wherein the sounds to remove are body generated sounds generated by a body of a user of the wireless earpiece.
2. The wireless earpiece of claim 1, wherein the removing the body generated sounds is performed by superimposing a cancellation signal to the audio.
3. The wireless earpiece of claim 1, wherein the removing the body generated sounds is performed

by filtering the body generated sounds from the audio.

4. The wireless earpiece of claim 1, wherein the body generated sounds comprise at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

5. The wireless earpiece of claim 1, wherein the body generated sounds comprise sounds generated by movement of the user.

6. The wireless earpiece of claim 1, wherein the body generated sound comprises sounds generated by items worn by the user as the user moves.

7. The wireless earpiece of claim 1 wherein the processor is further configured to receive the audio from one or more microphones of a second wireless earpieces in operative communication with the wireless earpiece.

8. A method of removing body generated sounds by an earpiece comprising: providing the earpiece, the earpiece comprising a wireless earpiece housing, a processor disposed within the wireless earpiece housing, a plurality of microphones operatively connected to the processor and at least one speaker operatively connected to the processor, wherein the processor is configured to receive audio from the plurality of microphones, perform processing on the audio to provide processed audio, and output the processed audio to the at least one speaker; receiving audio from the plurality of microphones at the processor; performing processing on the audio to provide processed audio, wherein the processing of the audio comprises identifying body generated sounds generated by a body of a user of the wireless earpiece and removing the body generated sounds from environmental audio detected by the plurality of microphones based on training data associated with body generated sounds; and outputting the processed audio to the at least one speaker of the wireless earpiece.

9. The method of claim 8 wherein the plurality of microphones include an air microphone and a bone microphone.

10. The method of claim 8 wherein the removing the body generated sounds is performed using active noise cancellation.

11. The method of claim 8 wherein the removing the body generated sounds is performed by filtering.

12. The method of claim 8 wherein the body generated sounds comprise at least one of a click or a pop generated by movement of a temporomandibular joint of the user.

13. The method of claim 8 wherein the body generated sounds comprise sounds generated by movement of the user.

14. The method of claim 8 wherein the body generated sound comprises sounds generated by items worn by the user as the user moves.

15. The wireless earpiece of claim 8 wherein the processor is further configured to receive the audio from one or more microphones of a second wireless earpieces in operative communication with the wireless earpiece.

16. A wireless earpiece comprising: a wireless earpiece housing; a processor disposed within the wireless earpiece housing; a plurality of microphones operatively connected to the processor; at least one speaker operatively connected to the processor; wherein the processor is configured to receive audio from the plurality of microphones, perform processing of the audio to provide processed audio, and output the processed audio to the at least one speaker to provide an audio transparency function; wherein the processing of the audio comprises identifying body generated sounds generated by a body of a user of the wireless earpiece to remove based on training data associated with sounds to remove and then remove the sounds to remove from the processed audio.
